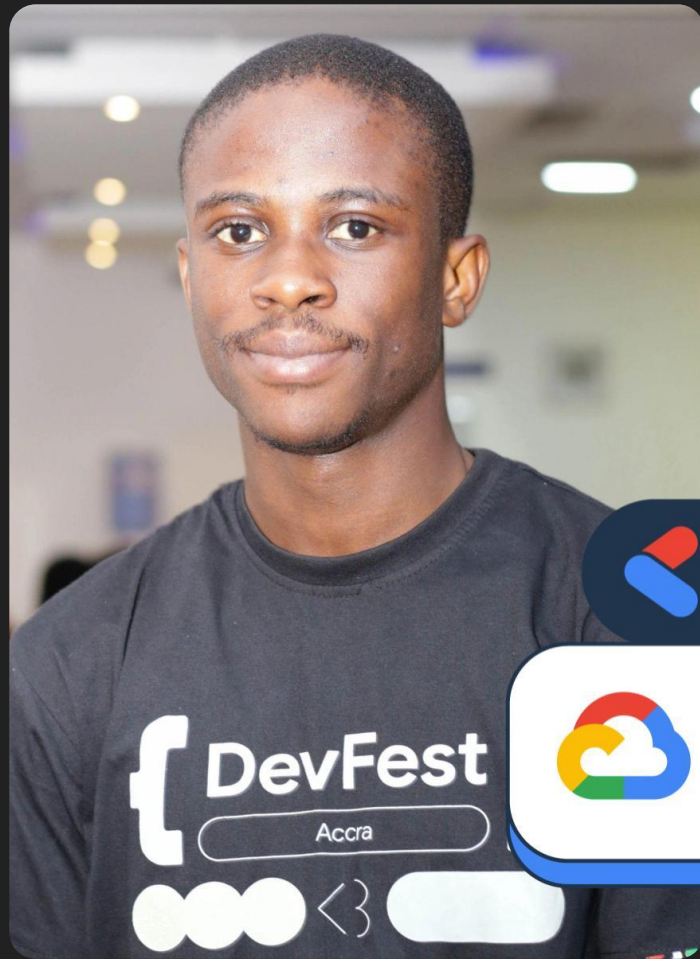


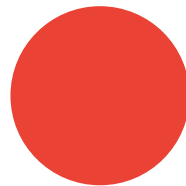
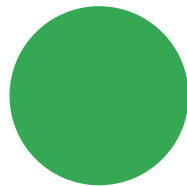
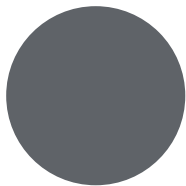
# Building a World Cup Analytics Agent using the Managed Agents API on Agent Platform

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# Evolution of AI Use Cases



## Predictive AI

- Regression & Classification
- Forecasting
- Sentiment Analysis
- Entity Extraction
- Object Detection

## Generative AI

- Text, Image & Code Gen
- Rewriting & Formatting
- Summarization
- Extractive Q&A
- Descriptions

## Multimodal Generative AI

- Image Understanding
- Spatial Reasoning
- Math in Context
- Video Q&A
- Speech & Translation

## Agentic AI

- Autonomous Agents
- Goal-Oriented Tasks
- Tool Use & Planning
- Multi-Agent Collaboration
- Self-Correction

# Why AI Agents?



Access external  
tools



Retrieve real-time  
info



Work with files &  
repos



Maintain conversation history



Execute multi-step tasks

Traditional chatbots struggle with these requirements. **AI Agents bridge this gap.**



# What is the Managed Agents API?

The Managed Agents API provides a fully managed way to build and deploy AI agents.



Agent lifecycle management



Tool integration



Persistent environments



Multi-turn conversations



Long-running workflows



Enterprise-ready  
deployment

Instead of managing complex agent infrastructure yourself, **the Agent Platform handles it for you.**  
Powered by the **Antigravity Harness**.

# Antigravity 2.0 Harness

The engine powering low-latency, scalable agentic workflows with enhanced orchestration.



Ultra-Low Latency Execution



Dynamic Path Routing



Isolate Sandbox Security



Stateful Context Memory



Async Event Propagation



Real-time Telemetry

Antigravity 2.0 provides the **computational backbone** for the Managed Agents API, ensuring reliability at enterprise scale.

# ◆ Gemini Enterprise Agent Platform

## Build

Agent Development Kit

New

3P agent frameworks

Agent Studio

New

Agent Garden

### Gemini API and Model Garden

Gemini models

3P and open models

Model training

Model inference

### Tools, data, and other agents

A2A

Grounding

RAG

MCP

Search

APIs and connectors

A2UI

AP2 and UCP

Cloud Marketplace

## Scale

Agent Runtime

GA

Agent Sessions

GA

Agent Sandbox

GA

Agent Memory Bank

GA

## Govern

Agent Gateway

New

Agent Identity

GA

Agent Registry

New

Agent Anomaly Detection

New

Model Armor

Agent Policy

Agent Security

New

Agent Compliance

## Optimize

Agent Evaluation

New

Agent Simulation

New

Agent Observability

New

Agent Optimizer

New

# System Architecture: Managed Agents API

Managed Agents API on Agent Platform leverages two key interfaces:

## CONTROL PLANE



### Agents API

Manage the lifecycle of your agents. Use this interface to create agents from configurations and set up execution environments.

- Mount external data sources
- Configure network allowlists
- Define sandbox environments

## DATA PLANE

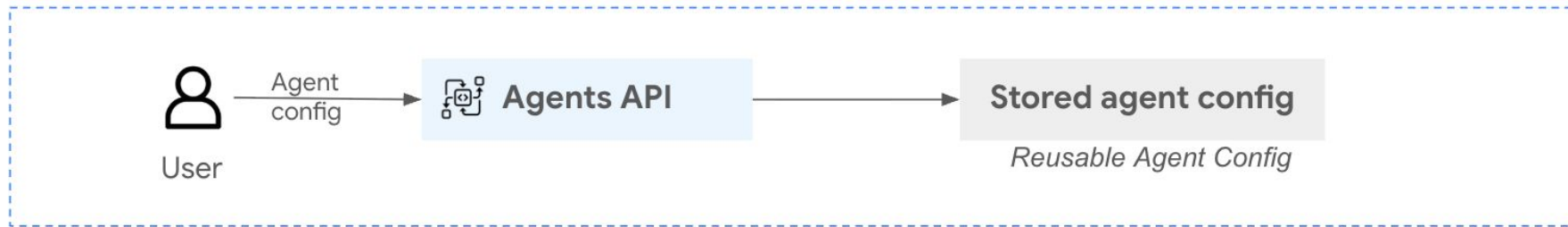


### Interactions API

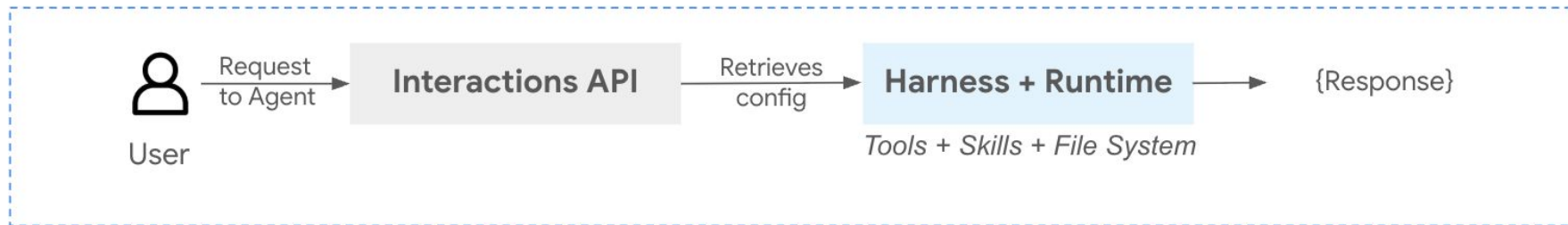
The primary interface for real-time interaction. It connects directly to the agents deployed via the Agents API.

- Runtime communication
- Bi-directional message flow
- Agent status monitoring

## The Agent Setup



## The Usage



**System architecture of Managed Agents API on Agent Platform.**



# World Cup Analytics Use Case

## OUR GOAL

Build an AI agent for comprehensive tournament intelligence:



### Analyze & Generate Insights

Historical data analysis and statistics generation



### Real-time Search

Search for current tournament information



### Contextual Conversations

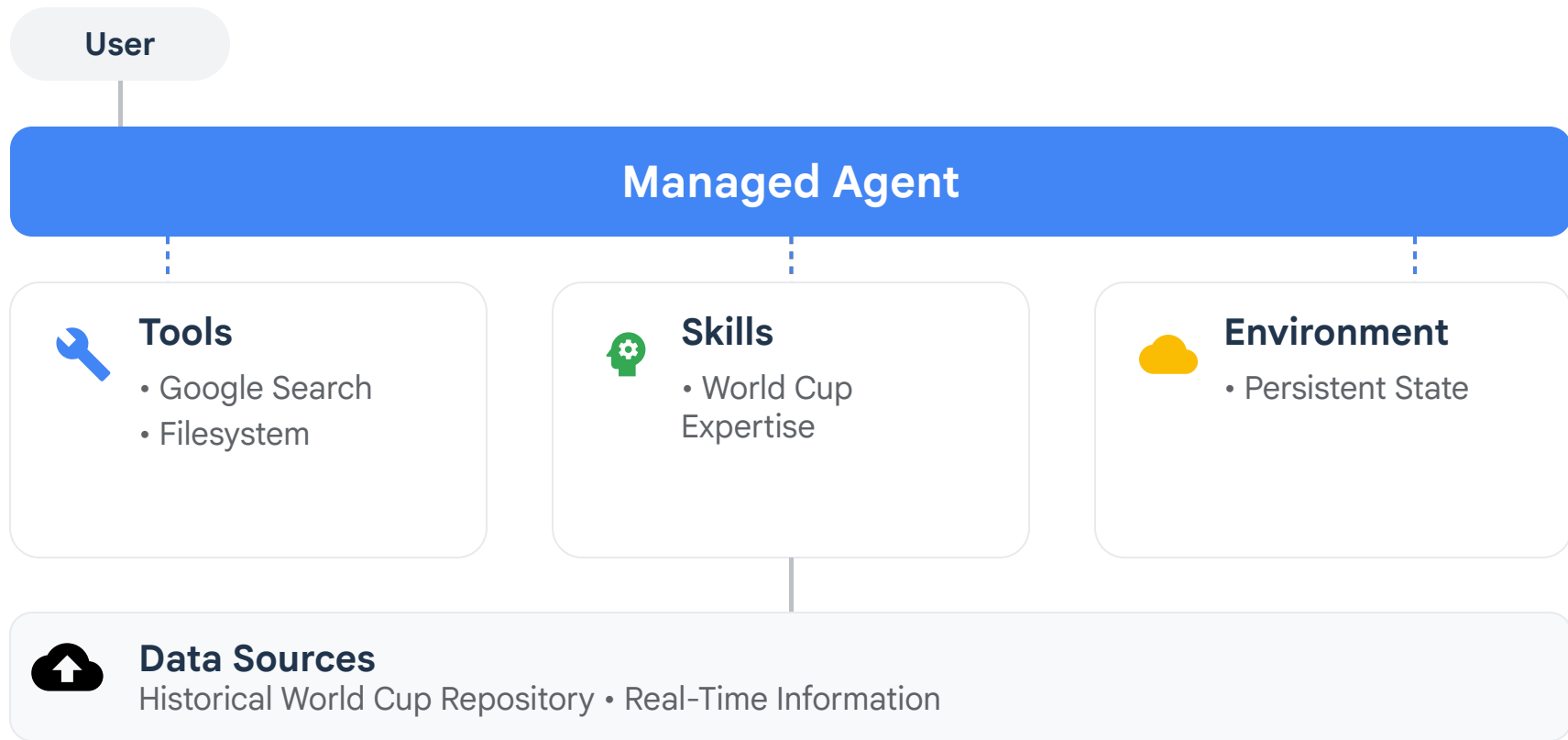
Multi-turn dialogue with context maintenance

## Cross-Domain Versatility

The same modular architecture can be seamlessly transitioned to other data-intensive fields:

- Research & Analytics
- Finance & Market Intelligence
- Healthcare Systems
- Customer Support

# Architecture Overview



# The SKILLS.md Framework



## Standard Definition

SKILLS.md serves as the primary instruction file for extending agent capabilities using best practices.



## Agent-Triggered

Unlike workflows, skills are triggered by the agent when it identifies relevance to the current task.



## Modular Structure

Defined within .agent/skills/ with optional scripts, examples, and resource sub-folders.



## Rules vs. Workflows vs. Skills

- Rules: System instructions (Always-on)
- Workflows: Saved prompts (User-triggered)
- Skills: Saved prompts & scripts (Agent-triggered)

# The KNOWLEDGE.md Standard



## Documented Wisdom

KNOWLEDGE.md centralizes domain-specific information, acting as the agent's long-term semantic memory.



## Contextual Retrieval

Provides high-density context for RAG pipelines, ensuring the agent stays grounded in factual source material.



## Structured Format

Uses Markdown for clean hierarchy, allowing agents to navigate complex datasets using headers and tables.



## Knowledge Management Strategies

- Curation: Keep content updated to prevent agent hallucination.
- Granularity: Break down large topics into digestible Markdown sections.
- Linking: Reference external repositories for deeper technical specifications.

# What We'll Build Today

During the codelab we will execute these key steps:

01 Configure a managed agent

02 Deploy the agent

03 Add tools and skills

04 Create a persistent environment

05 Enable multi-turn conversations

06 Analyze World Cup data

Let's build it.



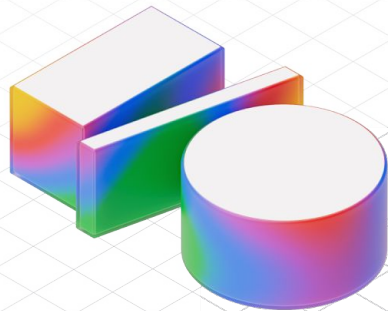
# CODELAB

[bit.ly/managed-agents-api](https://bit.ly/managed-agents-api)



# Additional Resources

- Managed Agents API on Agent Platform sandbox environment:  
<https://docs.cloud.google.com/gemini-enterprise-agent-platform/build/managed-agents/sandbox-environment>
- Create and manage agents:  
<https://docs.cloud.google.com/gemini-enterprise-agent-platform/build/managed-agents/create-manage>
- Managed Agents API Documentation:  
<https://docs.cloud.google.com/gemini-enterprise-agent-platform/build/managed-agents>
- Gemini Enterprise Agent Platform:  
<https://docs.cloud.google.com/gemini-enterprise-agent-platform>



# QUESTIONS





# Asante Sana!



@gabbyagbobli



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